Constructing a 0+1 field theory: the quartic oscillator and the Yang–Mills matrix model

Result, 2026-10-27; standard construction assembled here, not a novel existence theorem. For any supplied $m, \lambda, \hbar > 0$, the one-component ϕ_1^4 theory has a continuous, stationary Euclidean field on the full time line, a reflection-positive bounded-cylinder algebra, a non-Gaussian vacuum, and a reconstructed positive-energy quantum system with a unique vacuum and a positive spectral gap. The finite-time lattice measures with ground-state boundary weights converge under arbitrary time refinement. Their real-time counterpart is the nonrelativistic quartic oscillator. The same construction applies to the existing $SU(2)$ bosonic matrix Hamiltonian, including its gauge-singlet sector, using the compact-resolvent result already proved in low-dimensional mass gap, Theorem 6.

The one-dimensional Euclidean field has only time as a base-space coordinate, so spacelike locality and Lorentz boosts impose no nontrivial condition. A relativistic particle may instead be described by a one-dimensional worldline theory with a spatial target and a different Hamiltonian. Neither construction derives $\hbar > 0$, proves the original 1998 consistency conjecture, or constructs four-dimensional Yang–Mills. The statements below use established Feynman–Kac, semigroup and spectral theorems; the assembly is written here, without a separate referee check.

1. Model and prior work

Put

$$V(q) = \frac{\lambda}{4} q^4, \qquad S_E[q] = \int \left(\frac{m}{2} \dot{q}^{\,2} + V(q) \right) d\tau. \tag{1}$$

The symbolic $e^{-S_E/\hbar} \mathcal{D}q$ below means an actual Brownian-bridge tilt or its stationary extension, not a Lebesgue measure on path space. Brownian paths are almost surely nowhere differentiable; the kinetic integral in (1) is therefore a mnemonic for the reference measure, not an ordinary pathwise integral.

The Rivero 1998 test, Theorems 2(b) and 3, already proves the fixed-$\hbar$ real-time product limit and the zero-endpoint Euclidean bridge limit for the potential $\lambda_{\text{old}} q^4$; here $\lambda_{\text{old}} = \lambda/4$. Its zero-endpoint bound also permits $\hbar \downarrow 0$, so it does not select a quantum action unit. The nonlinear cell is one exact elimination step for (1), with its full normalized interaction retained. Its one-step estimates alone were not a stationary continuum construction.

Feynman–Kac and the $P(\phi)_1$ process are established literature: Simon 1999 (the bounded-below case cited and its theorem passage read in the 1998 test), and Rosen–Simon 1976 (bibliographic record). The real-time quartic oscillator also has an oscillatory-integral treatment by Albeverio–Mazzucchi 2005 (abstract). None of these references is a novelty claim.

2. A vacuum and a gap from the Hamiltonian

Define on $L^2(\mathbb{R}, dq)$ the Friedrichs operator

$$H = -\frac{\hbar^2}{2m} \frac{d^2}{dq^2} + \frac{\lambda}{4} q^4. \tag{2}$$

The form domain is $H^1(\mathbb{R}) \cap L^2(\mathbb{R}, q^4 dq)$. The form is closed and nonnegative. It has compact embedding into L^2: Rellich compactness holds on every bounded interval, while its unit ball satisfies

$$\int_{|q|>R} |\psi(q)|^2 dq \le R^{-4} \int q^4 |\psi(q)|^2 dq. \tag{3}$$

Thus H has compact resolvent. Feynman–Kac makes $e^{-tH/\hbar}$ positivity improving for $t > 0$: the free heat kernel is strictly positive and every finite bridge has a strictly positive potential weight.

By the positivity-improving ground-state theorem, the bottom eigenvalue E_0 is simple and its normalized eigenfunction Ω is strictly positive. Discreteness implies $E_1 > E_0$ and

$$\Delta = E_1 - E_0 > 0. \tag{4}$$

Writing $\ell = (\hbar^2/(m\lambda))^{1/6}$ and $q = \ell x$, unitary rescaling gives

$$H = \hbar^{4/3}\lambda^{1/3}m^{-2/3}\left(-\tfrac{1}{2}\frac{d^2}{dx^2} + \tfrac{1}{4}x^4\right) \tag{5}$$

up to that unitary conjugation. Thus the gap is a positive dimensionless eigenvalue difference times $\hbar^{4/3}\lambda^{1/3}m^{-2/3}$; it vanishes in the indicated small-$\hbar$ or small-λ limits. This does not imply positivity of $\hbar$ from classical premises.

3. Arbitrary time refinement

For $s > 0$, let

$$p_s(x,y) = \left(\frac{m}{2\pi\hbar s}\right)^{1/2} e^{-m(x-y)^2/(2\hbar s)}$$

and define the symmetric single-step transfer kernel

$$K_s^{\mathrm{lat}}(x,y) = p_s(x,y)e^{-s[V(x)+V(y)]/(2\hbar)}. \tag{6}$$

For a partition π of $[0,T]$, their product and integration over internal nodes equal $p_T(x,y)$ times the expectation, under a Brownian bridge of variance parameter $\hbar/m$ from x to y, of $\exp[-W_\pi/\hbar]$, where

$$W_\pi = \sum_j \frac{t_{j+1}-t_j}{2}[V(q(t_j)) + V(q(t_{j+1}))]. \tag{7}$$

The bridge is continuous almost surely, so its polygonal interpolation converges uniformly and $W_\pi \to \int_0^T V(q(t))dt$ for every sequence with $|\pi| \to 0$. Since $0 \le e^{-W_\pi/\hbar} \le 1$, dominated convergence gives the Feynman–Kac bridge, including bounded continuous path observables. This is the argument of the 1998 test, extended from zero to arbitrary fixed endpoints; no change of $\hbar$ accompanies the refinement.

To obtain vacuum rather than fixed-endpoint data, weight the two ends by $\Omega(x)\Omega(y)\,dx\,dy$ and normalize. The free heat kernel bounds the tilted kernels, and the L^2 contraction of the free semigroup gives an integrable bound against $\Omega(x)\Omega(y)$. Dominated convergence therefore applies to these finite-interval distributions. Their continuum normalizer is $\langle\Omega, e^{-TH/\hbar}\Omega\rangle = e^{-TE_0/\hbar}$. Ground-state boundary weights make the continuum restrictions consistent as the interval grows. Kolmogorov consistency and the locally continuous bridge laws give a stationary process $q(\tau)$ on the full time line. This construction uses the continuum ground state at the boundary; it does not assert uniform interchange of lattice-spacing and infinite-time limits for arbitrary lattice boundary states.

Equivalently, with $A = H - E_0$ and kernel K_t of $e^{-tH/\hbar}$, the stationary one-time law and transition are

$$\mu(dq) = \Omega(q)^2 dq, \qquad P_t(x,dy) = e^{tE_0/\hbar}\frac{\Omega(y)}{\Omega(x)}K_t(x,y)dy. \tag{8}$$

The eigenvalue equation makes P_t Markov; symmetry of K_t makes it reversible for μ. Times $\tau_1 \le \cdots \le \tau_n$ and bounded functions f_j give

$$\mathbb{E}\left[\prod_{j=1}^n f_j(q(\tau_j))\right] = \langle\Omega, f_1(Q)e^{-(\tau_2-\tau_1)A/\hbar}f_2(Q)\cdots f_n(Q)\Omega\rangle. \tag{9}$$

4. Reflection positivity and non-Gaussianity

Let F be a finite linear combination of products of bounded functions of $q(t)$ at nonnegative times, and let Θ reflect time and complex-conjugate the observable. From (9),

$$\mathbb{E}[(\Theta F)F] = \|v_F\|^2 \geq 0, \tag{10}$$

where for a single term with $0 \leq t_1 \leq \cdots \leq t_n$,

$$v_F = e^{-t_1 A/\hbar} f_1(Q) e^{-(t_2-t_1)A/\hbar} \cdots f_n(Q)\Omega,$$

and v_F is extended linearly. This is reflection positivity for the bounded cylinder algebra. The associated reconstruction gives the semigroup $e^{-tA/\hbar}$ and Hilbert space $L^2(\mathbb{R})$: the vectors $f(Q)\Omega$, for bounded f, are dense because $\Omega > 0$ almost everywhere. The coordinate Q is self-adjoint as multiplication by q and $e^{-itA/\hbar}$ gives real-time unitary dynamics.

The potential is even, so uniqueness and positivity make Ω even. The field is not Gaussian. If the one-time density Ω^2 were Gaussian, positivity would make $\Omega = Ce^{-\alpha q^2/2}$, $\alpha > 0$. But substituting this into $H\Omega = E_0\Omega$ leaves the nonzero quartic term $\lambda q^4/4$, which cannot be cancelled by the constant and quadratic terms from the derivative. Thus even the one-time marginal is non-Gaussian. This is a continuum witness, unlike the unscaled shrinking cell cumulant in the Alpoge–Balaban peek.

For bounded real f, g, spectral projection also gives

$$|\operatorname{Cov}_\mu(f(q(0)), g(q(t)))| \leq e^{-t\Delta/\hbar} \|f\Omega\| \, \|g\Omega\|. \tag{11}$$

The gap is an energy gap of this explicitly constructed quantum mechanical Hamiltonian, not a four-dimensional field-theory mass gap. The claim here is the bounded-cylinder reconstruction; it does not separately verify a chosen full hierarchy of unbounded-field growth axioms.

5. The gauge-theory rung and relativity

A literal pure gauge connection on a one-dimensional base has no two-form curvature: $F_{00} = 0$. A time circle can retain a global holonomy. The nontrivial model connected to higher-dimensional Yang–Mills is instead its dimensional reduction: spatial gauge components survive as adjoint matrices $X_i(t)$ and the temporal gauge field imposes Gauss's law. For $SU(2)$ and $D \geq 2$ matrices, the matrix theorem proves compact resolvent, a simple positive vacuum, and a gap on the full configuration space and its gauge-invariant subspace. Its potential is a sum of squared commutators and has unbounded commuting valleys, so the scalar argument $V \to \infty$ cannot be reused. The multicomponent version of (6)–(10), restricted to bounded invariant functions, supplies the stationary reflection-positive gauge-singlet construction; positivity and simplicity make the ground state invariant. This finite-dimensional matrix model keeps only spatially constant modes; it does not justify discarding the other modes of the full theory or construct Yang–Mills on $\mathbb{R}^4$. No $SU(3)$ matrix gap is asserted here.

The field theory (1) is a scalar theory on a one-dimensional spacetime, and is ordinary nonrelativistic quantum mechanics when q is read as the coordinate of a particle. There are no nontrivial boosts in $0+1$ spacetime dimensions. A relativistic particle in d target-space dimensions instead can use the positive-energy Hamiltonian

$$H_{\text{rel}} = \sqrt{c^2P^2 + m^2c^4} - mc^2 + V(Q). \tag{12}$$

Self-adjointness gives a unitary time evolution when the operator is well-defined, and the positive semigroup can have a subordinate-Brownian, jump-process representation under suitable potential assumptions. See Hiroshima–Ichinose–Lorinczi (abstract and introduction) and Gheryan et al. (abstract), which treats both relativistic and nonrelativistic Nelson models. Their theorems

have their own assumptions. A confining external V selects a frame; it does not establish full Poincare covariance. For $V = 0$ on $\mathbb{R}^d$, the bottom of (12) is not a normalizable vacuum, so the stationary ground-state construction above cannot be applied without additional structure. A proper-time worldline and the Klein–Gordon/Feynman propagator require their own constraint and normalization conventions.

6. Consequence for STATE

The preliminary $0 + 1$ existence statement is a standard theorem for specified quantum weights and is now assembled with vacuum, reflection positivity, non-Gaussianity and the continuum limit. The same architecture connects to the existing $SU(2)$ matrix-model gap. The open Newton question remains whether independently justified premises require a positive universal action scale; the $\hbar = 0$ alternatives and the real-time/classical distinction remain essential.